

Distributed Data Mining

Introduction

- As the amount of data collected grows, most businesses and organizations are turning to Data Mining for analysis.
- Data Mining helps identify patterns and discover trends that further help in making right decisions for the company and its growth.
- The Data Mining method is applied to the acquired data in order to derive model predictions and find interesting trends. Here, most of the data is stored on a single site.
- But, there are many applications where the data is inherently distributed.
- With the developments in Data Mining, the concept of Distributed Data Mining (DDM) came into action.
- Distributed Data Mining involves the mining of datasets regardless of their physical locations.
- Its main role is to extract information from the distributed heterogeneous Databases and use it for decision-making

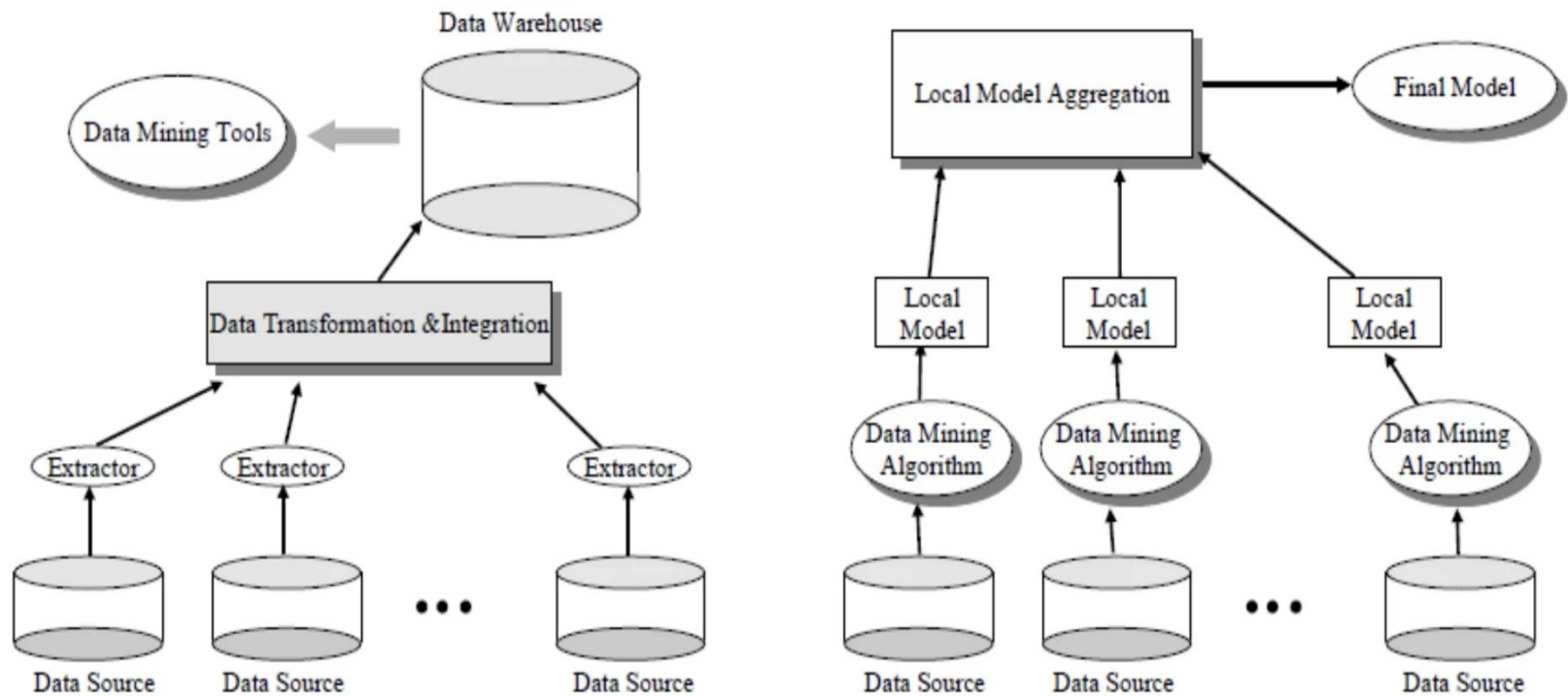
Introduction

- Data Mining is a process of sorting large datasets to discover valuable information and use it for analysis to increase efficiency in business operations.
- It uses software, algorithms, and other statistical methods to identify patterns and relationships that further help in resolving business issues.
- Data mining processes are mostly used in marketing, risk management, cyber security, mathematics, medical diagnosis, and other fields.
- To uncover trends and handle business difficulties, most companies employ Data Mining to extract hidden patterns and information from huge datasets.
- It includes machine learning and statistical analysis in addition to data management activities.
- Organizations may use Data Mining to discover possible customer service issues, enhance lead conversion rates, recognize market trends and estimate product demand, better analyze cyber security and other risks, and decrease redundancy, among other things.

What is Distributed Data Mining?

- Distributed Data Mining process involves the mining of distributed datasets stored in multiple local Databases.
- Often the data is distributed among several Databases, which makes it more prone to security risks.
- With the help of Distributed Data Mining, admins can perform data analysis and mining operations in a distributed manner to discover knowledge and use it efficiently for business operations.

Architecture of Distributed Data Mining (DDM)



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- In the architecture, the left section shows that the Data Mining has a traditional Data Warehouse-based architecture. Here, for centralized Data Mining, the admin needs to upload critical data to the Data Warehouse.
- However, this architecture is not supported by Distributed Data Mining because it lacks proper use of distributed resources, supports long response time, and comprises characteristics of a centralized Data Mining algorithm.
- The only solution to this is to set up a distributed application for processing controlled by potential resources and human factors.
- DDM performs all the Data Mining operations on the basis of available resources and types of operations. It picks the site to access data based on storage, computing, and connection capabilities and then conducts all processes centrally.

What are the Processes in DDM?

Before the mining process begins, the data is prepared by selecting the appropriate information, eliminating noisy data, and integrating data from multiple Databases. Data Cleansing, Integration, Reduction, Transformation, Mining, Pattern Evaluation, and Knowledge Representation are all components of the Data Mining process.

1) Data Cleaning

Data Cleaning is the primary step under which all the noisy, inconsistent, or incomplete data is removed from the collection. Simply, it removes any noisy data that isn't required for the analysis.

2) Data Integration

As part of the Data Integration process, all information that comes from different datasets, including databases, data cubes, data warehouses, or files are combined to perform analysis. This step aids in improving the efficiency and speed of the data mining process.

3) Data Reduction

This technique helps in sorting and obtaining only relevant data from the collection for analysis. It focuses on reducing the number of attributes and original data volume while maintaining integrity.

What are the Processes in DDM?

4) Data Transformation

For the Data Mining process, the data is aggregated and converted here. As a result, understanding and identifying trends in the mining process is simplified.

5) Data Mining

Under the Data Mining process, experts look for new patterns and try gathering information from large datasets to perform analysis and resolve business issues.

6) Pattern Evaluation

Based on the interestingness measures, some interesting patterns are identified. After identifying patterns, Data Summarization and Visualization techniques are practiced to evaluate and make it easier for the user to understand the data.

7) Knowledge Representation

Here all the mined information and data is visualized in the form of reports and presented using knowledge representation tools to the user.

DDM Algorithms: 3 Key Types

Distributed Data Mining Algorithms can be classified as:

- 1. Multi-Agent System:** The Multi-Agent System (MAS) algorithm is mostly used in cases where there is a need to compare data at different nodes.
The behavior of agents in a Multi-Agent System (MAS) depends completely on the data collected from distributed sources. This mechanism is beneficial for DDM as all the agents are identical and interact in a shared environment to solve problems.
- 2. Meta-learning:** Implemented by the JAM system, the Meta-learning algorithm is a technique in which local classifiers or models are generated from distributed datasets. These classifiers are later used to produce global classifiers.
Basically, under this algorithm DDM performs partial analysis at different locations and forwards a summarized version of the analysis to peer sites for further analysis.
- 3. Grid:** This algorithm allows organizations to distribute compute-intensive data among remote resources and mine data where it is stored.

Benefits of DDM

Initially, Data Mining was limited to sorting centralized datasets stored at a single site. But with more and more usage of data, multiple interrelated Databases were created & distributed over a large computer network.

The Data Mining technique is incapable of dealing with distributed datasets. As a result, the concept of distributed data mining was introduced. Here are a few benefits of distributed data mining:

- There are many multinational companies (MNCs) where the data is inherently distributed. Sending all the data to a central site for data mining is a great solution, but it can be a time-consuming and expensive process because of its large size. In this case, using Distributed Data Mining process is the best solution.
- Distributed Data Mining can handle large datasets that are beyond the Data mining capability and at a faster pace as they distribute the workload among different sites.
- Distributed Data Mining allows the execution of multiple queries at different sites at the same time, which leads to improvement in performance.
- The technique delivers faster results which further aids businesses in planning strategies and managing operations.
- It helps create analytical models and insightful reports that help businesses in making better decision-making.